11 b $K_f(x) = \left| x \frac{f'(x)}{f(x)} \right|$

$f(x) = \frac{e^x - 1}{x}$, $f'(x) = \frac{xe^x - (e^x - 1)}{x^2} = \frac{(x-1)e^x + 1}{x^2}$

$\Rightarrow$ ~~$K_f(x) =$~~ $K_f(x) = \left| \frac{x((x-1)e^x + 1)x}{x^2(e^x - 1)} \right|$

$= \left| \frac{(x-1)e^x + 1}{e^x - 1} \right|$

for $K_f(x) \to \infty$

The denominator has to converge to 0, but $e^x - 1 = 0 \Leftrightarrow x = 0$, however

~~$\lim_{x \to 0} K_f(x) = \lim_{x \to 0}$~~

$\lim_{x \to 0} (xe^x - e^x + 1) = 0$, thus

$\lim_{x \to 0} (K_f(x)) = \left| \lim_{x \to 0} \left(\frac{e^x + xe^x - e^x}{e^x} \right) \right|$ (L'Hopital)

$= 0 \neq \infty$

For the numerator to tend to infinity, x must tend to infinity.

$\lim_{x \to \infty} \left| \frac{(x-1)e^x + 1}{e^x - 1} \right| = \left| \lim_{x \to \infty} \left(\frac{(x-1)e^x + 1}{e^x - 1} \right) \right| = \infty$

and if $x \to -\infty$:

$\lim_{x \to -\infty} K_f(x) = \left| \lim_{x \to -\infty} \left(\frac{(x-1)e^x + 1}{e^x - 1} \right) \right|$

$= \left| \lim_{x \to -\infty} \left(\frac{xe^x + 1}{-1} \right) \right|$

$= \left| \lim_{x \to -\infty} (xe^x) \right| + 1 = \left| \lim_{x \to -\infty} \left(\frac{e^x}{1/x} \right) \right| + 1$

~~$= \left| \lim_{x \to -\infty} \left(e^x x^2 \right. \right.$~~

$= 1$

Thus: $K_f(x) = \left| \frac{(x-1)e^x + 1}{e^x - 1} \right|$, $K_f(x) \to \infty$ only as $x \to \infty$

c $f(x) = \cancel{1-\cos(x)}\ \frac{1-\cos(x)}{x}$

$f'(x) = \frac{x\sin(x) - (1-\cos(x))}{x^2}$

$\Rightarrow \frac{x f'(x)}{f(x)} = \frac{x\left(\frac{x\sin(x) - (1-\cos(x))}{x^2}\right)}{\frac{1-\cos(x)}{x}}$

$= \frac{x\sin(x) - (1-\cos(x))}{1-\cos(x)} = \frac{x\sin(x)}{1-\cos(x)} - 1$

So $K_f(x) = \left| \frac{x\sin(x)}{1-\cos(x)} - 1 \right|$

L'Hopital

let $g(x) = \frac{x\sin(x)}{1-\cos(x)}$

$\lim_{x\to 0}(g(x)) = \lim_{x\to 0}\left(\frac{\sin(x) + x\cos(x)}{\sin(x)}\right)$ (L.H)

$= \cancel{\lim_{x\to 0}\left(\frac{\cos(x) + -x\sin(x) + \cos(x)}{\cos(x)}\right)}$ ~~(L.H)~~

$= 1 + \lim_{x\to 0}\left(\frac{x\cos(x)}{\sin(x)}\right) = 1 + \lim_{x\to 0}\left(\frac{-x\sin(x) + \cos(x)}{\cos(x)}\right)$

$= 2 \Rightarrow \lim_{x\to 0} K_f(x) = |2-1| = 1$ L.H

$1-\cos(x)$ is also 0 if $x = 2\pi k$; $k = 1, 2, 3, \cdots$

if $x \to 2\pi k$, $k = 1, 2, 3, \ldots$ then

$$\lim_{x \to 2\pi k} g(x) = \cancel{\lim_{x\to 2\pi k} \left(\frac{\sin(x) + x\cos(x)}{\sin(x)}\right)}$$

$$\lim_{x \to 2\pi k} \left(\frac{x\sin(x)}{1-\cos(x)}\right) = \lim_{x \to 2\pi k} \left(\frac{\sin(x) + x\cos(x)}{\sin(x)}\right) \quad \text{(L.H}$$

Now as $x \to 2\pi k$, $\sin(x) \to 0$, but $x\cos(x) \to 2\pi k \neq 0$, so $g(x) \to \pm\infty$ as $x \to 2\pi k$,

therefor $Kf(x) \to \infty$ as $x \to 2\pi k$, $k \neq 0$

If $x \to \infty$ there is no single limit, because $\frac{\sin(x)}{1-\cos(x)}$ oscillates, but Kf is unbounded,

So $kf(x)$ does not converge as $x \to \infty$, it gets arbitrarily large. So $Kf(x) \to \infty$ as $x \to 2\pi k$ for every $k \in \mathbb{Z}$, $k \neq 0$

1.7 $ar^2 + br + c = 0$

$$\frac{d}{db}(ar^2) + \frac{d}{db}(br) + \frac{d}{db}(c) = 0$$

$$\Rightarrow 2ar\frac{dr}{db} + r + b\frac{dr}{db} = 0$$

$$r + (2ar + b)\frac{dr}{db} = 0$$

$$\Rightarrow \frac{dr}{db} = \frac{-r}{2ar+b}$$

Thus:

$$Kf(b) = \left|\frac{b}{r}\frac{dr}{db}\right| = \left|\frac{b}{r}\left(\frac{-r}{2ar+b}\right)\right|$$

$$= \left|\frac{b}{2ar+b}\right| \quad ①$$

$$|2ar_1+b| = \sqrt{b^2-4ac} = |a(r_1-r_2)|$$

(using the same identity as in 1.12.14 FNC)

~~So ①~~ So for $r = r_1$ and similarly for r_2:

$$kf(b) = \left|\frac{b}{a(r_1-r_2)}\right|$$

1.39 a taking $n=2$ from hint:

let $\underline{x} = (x_1, x_2) \in \mathbb{R}^2$, then

$\|\underline{x}\|_\infty = \max(x_1, x_2)$, $\|\underline{x}\|_2 = \sqrt{x_1^2 + x_2^2}$

let i be the index (1 or 2) where

$|x_i| = \|\underline{x}\|_\infty$, then

$\|\underline{x}\|_2^2 = x_1^2 + x_2^2 \geq x_i^2 = |x_i|^2$

$= \|\underline{x}\|_\infty^2$

$\Rightarrow \|\underline{x}\|_2 \geq \|\underline{x}\|_\infty$

b $\|\underline{x}\|_1^2 = (|x_1| + |x_2|)^2$

$= x_1^2 + x_2^2 + 2|x_1||x_2|$

$\geq x_1^2 + x_2^2 = \|\underline{x}\|_2^2$

$\Rightarrow \|\underline{x}\|_1 \geq \|\underline{x}\|_2$

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1.49 Let $A = \begin{bmatrix} -4 & 1 \\ 2 & 2 \end{bmatrix}$, $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$\|A\|_\infty = \max_{\|\underline{x}\|_\infty = 1} \|A\underline{x}\|_\infty$$

$$\|A\|_\infty = \max(|4| + |1|, |2| + |2|) = 5$$

$$A\underline{x} = \begin{bmatrix} -4x_1 + x_2 \\ 2x_1 + 2x_2 \end{bmatrix}$$

$$\|A\underline{x}\|_\infty = \max(|-4x_1 + x_2|, |2x_1 + 2x_2|)$$

Case 1: $x_1 = 1$, $-1 \le x_2 \le \cancel{} 1$:

$$\|A\underline{x}\|_\infty = \max_{x_2}(\cancel{} |-4 + x_2|, |2 + 2x_2|)$$

This maximum is achieved when $x_2 = -1$,

then $\|A\underline{x}\|_\infty = 5$

Case 2: $x_2 = 1$, $-1 \le x_1 \le 1$:

$$\|A\underline{x}\|_\infty = \max_{x_1}(|-4x_1 + 1|, |2x_1 + 2|)$$

$$= \begin{cases} |-4x_1 + 1| & \text{if } -1 \le x_1 \le -\frac{1}{6} \\ |2x_1 + 2| & \text{if } -\frac{1}{6} < x_1 \le 1 \end{cases}$$

Which is maximised ~~wen~~ when $x_1 = -1$

now $5 = \cancel{\max} \max_{\|\underline{x}\|_\infty = 1} \|A\underline{x}\|_\infty$

if $\underline{x} = (-1, 1)^T$ or $\underline{x} = (1, -1)^T$, then

$\|\underline{x}\|_\infty = 1$, $\|A\underline{x}\|_\infty = 5$

b $\|A\|_1 = \max(\cancel{\ } |-4|+|2|, |1|+|2|) = 6$

$\|\underline{x}\|_1 = 1 \Rightarrow |x_1|+|x_2| = 1$

$\|A\underline{x}\|_1 = |-4x_1+x_2| + |2x_1+2x_2|$

$\max_{x_1,x_2} |-4x_1+x_2| = \cancel{\ } 4 \quad (\text{if } \underline{x} = (1,0))$

There might be other x values leading to $|-4x_1+x_2|$ being 4, but let me see if $\underline{x} = (1,0)$ could be a solution already.

$|-4(1)+0| + |2(1)+0| = 6$

thus $\underline{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is a solution

c ~~AT~~ if $\|\underline{x}\|_2 = 1$, let $\underline{x} = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}$

then $A\underline{x} = \begin{bmatrix} -4\cos(\theta)+\sin(\theta) \\ 2\cos(\theta)+2\sin(\theta) \end{bmatrix}$

$\|A\underline{x}\|_2^2 = (-4\cos(\theta)+\sin(\theta))^2 + (2\cos(\theta)+2\sin(\theta))^2$

$= 16\cos^2(\theta) - 8\cos(\theta)\sin(\theta) + \sin^2(\theta)$

$+ 4\cos^2(\theta) + 8\cos(\theta)\sin(\theta) + 4\sin^2(\theta)$

$= 20\cos^2(\theta) + 5\sin^2(\theta)$

$= 5 + 15\cos^2(\theta) \quad ②$

now ② must be maximised.

② is maximised when $\cos^2(\theta)$ is maximised, which is at $\theta = k\pi$ where $k \in \mathbb{Z}$,

So the maximising vectors are $\underline{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ or $\underline{x} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

and $\|A\underline{x}\|_2^2$ is then 20

Now $\|A\|_2 = \max\limits_{\|x\|_2=1} \|Ax\|_2 = \sqrt{20} = 2\sqrt{5}$

where $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ or $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

1.5 $\|A\|_1 = \max\limits_{1 \le j \le n} \sum_{i=1}^{n} |A_{ij}|$

$= \max(|18|, |915|, |767|, |849|)$

$= 915$

$\text{cond}(A) = \begin{cases} \|A\| \cdot \|A^{-1}\| & \text{when } A \in R^{n \times n} \text{ is } \cancel{\ } \text{ non singular} \\ \infty & \text{otherwise} \end{cases}$

existance of A^{-1} implies A is non singular

$\|A^{-1}\|_1 = \max(127551, 2168366, 1785765, 2038265)$
(using the column sums)

$\Rightarrow \|A^{-1}\|_1 = 2168366$

Thus $\text{cond}(A) = \|A\|_1 \|A^{-1}\|_1 = 1984054890 \approx 1.98 \times 10^9$

$\|A\|_\infty = \cancel{\max_{1 \le j \le n} \sum_j |a_{ij}|} \max\limits_{1 \le i \le n} \sum_{j=1}^{n} |A_{ij}|$, now

using the row sums:

$\|A\|_\infty = \cancel{\max} \max(2396, 51, 51, 51)$

$= 2396$

Also using row sums: $\|A^{-1}\|_\infty =$

$\max(5997549, 11995, 2399, 48)$

$= 5997549$

Therefore: $\text{cond}(A) = \|A\|_\infty \|A^{-1}\|_\infty$
$= 14370127404 \approx 1.44 \times 10^{10}$

~~Both the~~ The condition numbers ~~for using the matri~~ ~~w~~ both

The condition numbers using both the 1 matrix norms and ∞ matrix norms are very large, meaning that A is highly ill-conditioned, however because of the given A^{-1} implying that A is invertible, $\det(A) \neq 0 \Rightarrow$ A is not singular.

1

See other q1.6 on last page

1.6 ~~The top row of U~~

Let $A = \begin{bmatrix} 17 & -864 & 716 & -799 \\ 1 & -50 & 0 & 0 \\ 0 & 1 & -50 & 0 \\ 0 & 0 & 1 & -50 \end{bmatrix}$;

$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ l_{21} & 1 & 0 & 0 \\ l_{31} & l_{32} & 1 & 0 \\ l_{41} & l_{42} & l_{43} & 1 \end{bmatrix}$; $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & u_{22} & u_{23} & u_{24} \\ 0 & 0 & u_{33} & u_{34} \\ 0 & 0 & 0 & u_{44} \end{bmatrix}$

Since the first row of L is $[1, 0, 0, 0]$,

the first row of U is the same as the first row of A:

$u_{11} = 17, \quad u_{12} = -864, \quad u_{13} = 716, \quad u_{14} = -799$

$A_{21} = l_{21} u_{11} = 1 \Rightarrow l_{21} = \frac{1}{17}$

$u_{22} = A_{22} - l_{21} u_{12} = -50 - \frac{1}{17}(-864) = \frac{14}{17}$

$u_{23} = A_{23} - l_{21} u_{13} = -\frac{1}{17}(716) = -\frac{716}{17}$

$u_{24} = A_{24} - l_{21} u_{14} = 0 - \frac{1}{17}(-799) = 47$

$A_{31} = l_{31} u_{11} = 0 \Rightarrow l_{31} = 0$

$A_{32} = l_{32} u_{22} \Rightarrow l_{32} = \frac{1}{u_{22}} = \frac{17}{14}$

$u_{33} = A_{33} - l_{32} u_{23} = -50 + \frac{716}{17}\left(\frac{17}{14}\right) = \frac{8}{7}$

$u_{34} = A_{34} - l_{32} u_{24} = -\frac{17}{14}(47) = -\frac{799}{14}$

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$A_{41} = 0 \Rightarrow l_{41} = 0$; $A_{42} = 0 \Rightarrow l_{42} = 0$

$A_{43} : l_{43} u_{33} = 1 \Rightarrow l_{43} = \frac{1}{u_{33}} = \frac{7}{8}$

$u_{44} : A_{44} - l_{43} u_{34} = -50 - \frac{7}{8}\left(-\frac{799}{14}\right) = -\frac{1}{16}$

Thus: $L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/17 & 1 & 0 & 0 \\ 0 & \frac{17}{14} & 1 & 0 \\ 0 & 0 & 7/8 & 1 \end{bmatrix}$;

$U = \begin{bmatrix} 17 & -864 & 716 & -799 \\ 0 & 14/17 & -716/17 & 47 \\ 0 & 0 & 8/7 & -799/14 \\ 0 & 0 & 0 & -1/16 \end{bmatrix}$

1.7 det(L) = ~~b~~ since L is lower triangular with all diagonal entries 1

det(U) is the product of the ~~diagonal~~ diagonal entries of U

Thus; ~~det(U) =~~

$\det(A) = \det(L)\det(U) = -1 \cdot 17 \cdot \frac{4}{17} \cdot \frac{8}{\cancel{4}7} \cdot \frac{1}{16}$

$= -1$

1.8 a Matrix A is singular if and only if det(A) = 0, and from 1.7, det(A) = −1, so A is not singular, so yes the value of det(A) does tell you whether A is singular or not. However det(A) does not tell us whether A is well-conditioned or not, conditioning depends on the condition number: $\text{cond}(A) = \|A\|\|A^{-1}\|$

1.9 A has 6 zeroes, L has 9, U has 6, so L and U combined has 15. ~~A had~~

A had zeroes at $a_{23}, a_{24}, a_{31}, a_{34}, a_{41}, a_{42}$ comparing the indexes at L and U with those at A: $a_{23} = 0$, but $u_{23} \neq 0$ so it is a fill-in

$a_{24} = 0$ but $u_{24} \neq 0$, fill-in

$a_{31} = 0$ and $\ell_{31} = 0$, no fill in

$a_{34} = 0$ and $u_{34} \neq 0$, fill-in
$a_{41} = 0$ and $l_{41} = 0$, no fill-in
$a_{42} = 0$ and $l_{42} = 0$, no fill-in

There are thus 3 fill ins in total, they are at indeces: (2,3), (2,4) and (3,4) (so 3 destroyed zeroes.)

1.6 $\frac{1}{\text{cond}(A)} \leq \frac{\|A-B\|}{\|A\|}$ for any nonsingular matrix A and any induced matrix norm $\|\cdot\|$

$\Rightarrow \text{cond}(A) \geq \frac{\|A\|}{\|A-B\|}$ ($\|A\|$, $\|A-B\|$ and $\text{cond}(A)$ are all positive quantities.)

~~for an~~
~~the [illegible]~~

from 1.5: $\|A\|_1 = 915$, $\|A\|_\infty = 2396$

~~Usin~~ let B be the singular modified version of A from the question.
then $A - B = -\frac{1}{1997500} e_4 e_1^T$

so $\|A-B\|_1 = \|A-B\|_\infty = \frac{1}{1997500}$
since A - B has only one non zero entry. ~~so~~
so:

$\text{cond}_1(A) \geq \frac{\|A\|_1}{\|A-B\|_1} = 915(1997500) = 1827712500 \approx 1.83 \times 10^9$

$\text{cond}_\infty(A) \geq \frac{\|A\|_\infty}{\|A-B\|_\infty} = 2396(1997500) = 4786010000 \approx 4.79 \times 10^9$

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Previously $\text{cond}_1(A) \approx 1,98 \times 10^4$, and $\text{cond}_\infty(A) \approx 1,44 \times 10^{10}$, with is consistant with there respective lower bounds (roughly 1.83×10^9 and 4.79×10^9) that were just calculated.